

WASP-43b

R-2623

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_140127 · created 2026-08-28T23:39:59+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456684.81647 +/- 0.00071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.053
Transit depth (Rp/Rs)^2	2.86 +/- 1.78 [%]
Orbital Inclination (inc)	78.97 +/- 2.88
Airmass coefficient 1 (a1)	1.023 +/- 0.068
Airmass coefficient 2 (a2)	-0.013 +/- 0.03
Scatter in the residuals of the lightcurve fit is	6.69 %
Best Comparison Star	#2 - [24, 105]
Optimal Method	PSF photometry
Transit Duration (day)	0.034 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.169 ± 0.053 vs 0.1594 (deviation 0.16σ)
Duration fit vs published	0.034 d vs 0.0512 d (deviation 0.82σ)
Mid-time O–C	1.6 min
Depth SNR	2.7
Residual scatter	6.69 %

Light curve

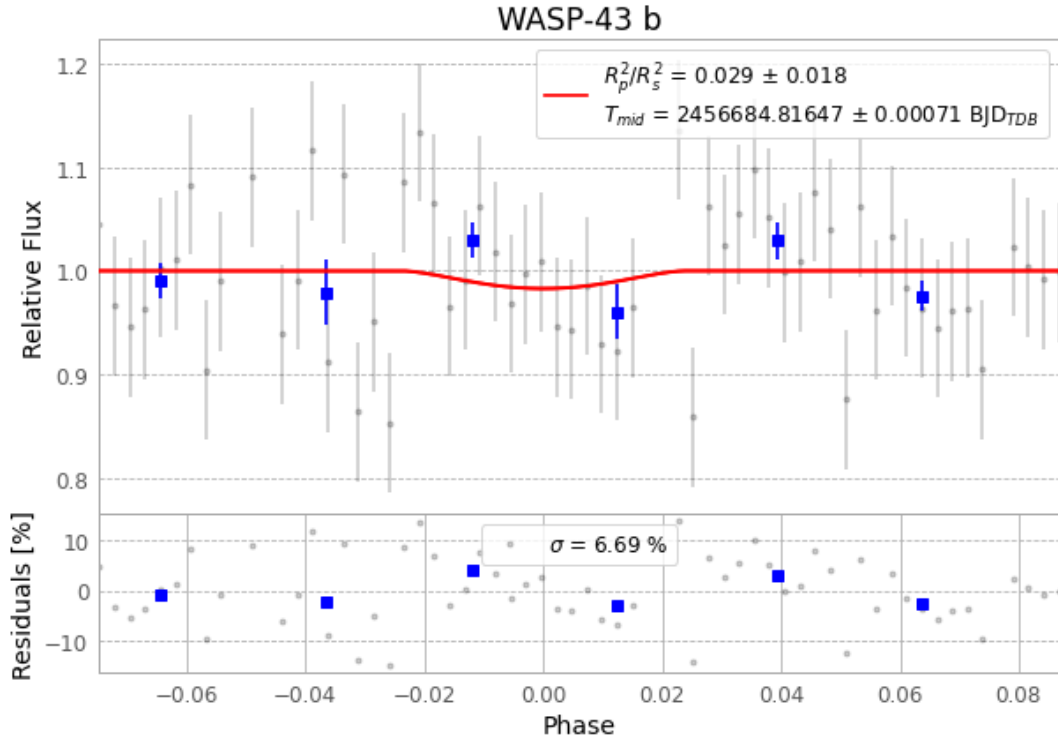


Figure 1 — FinalLightCurve_WASP-43 b_2014-01-26.png (SHA-256 6569b0f325571e84...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [585, 129], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 f5041114bd6fb33e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

62 FITS frames (41 MB), WASP-43140127060012.FITS → WASP-43140127090912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140127.log (5c5b8dc94990...), FinalLightCurve_WASP-43 b_2014-01-26.png (6569b0f32557...), FinalLightCurve_WASP-43 b_2014-01-26.csv (41f2596773b6...)

Compute resources

Wall time 107.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 23:40Z.